Oil production — Solution

X24 (2024) — English translation

A1 — 0.30 pts

Let the oil reservoir be a section of ancient river sandstone deposits in the shape of a parallelepiped with a height of $h = 10$ m, a width of $b = 100$ m and a length of $L = 2000$ m. Rock porosity $\varphi = 0.1$. Estimate the oil reserves m_n in this field. Express the answer in terms of L, b, h, ρ and φ, and also give its numerical value in tons. Consider that the oil fluid completely fills the pore volume.

From the definition of porosity for the volume of oil we find:

$$V_{\mathrm{o}} = \varphi \cdot bhL.$$

Since $m_{\mathrm{n}} = \rho V_{\mathrm{n}}$, we have:

Answer:

$$m_{\mathrm{o}} = \rho\varphi bhL = 160 \cdot 10^3 \text{ tons}.$$

A2 — 0.30 pts

Let the reservoir oil pressure at the bottom of the deposits be $p_r = 250$ atm. Find at what maximum depth H_{max} the deposit will be gushing, i.e. oil will flow to the surface under its own pressure. Express the answer in terms of ρ, g and p_r, and also give its numerical value. The compressibility of oil can be neglected.

The liquid will flow out under the influence of reservoir pressure if it exceeds the pressure of the hydrostatic column of oil:

$$p_{\mathrm{m.p.}} \geq \rho g h.$$

Thus:

Answer:

$$H_{max} = \frac{p_{\mathrm{m.p.}}}{\rho g} \approx 3.125 \text{ km}$$

A3 — 0.60 pts

Estimate the maximum possible oil recovery factor α_{max} in the flowing mode at reservoir pressure $p_r = 250$ atm, if the oil compressibility is $\beta = 5 \cdot 10^{-10}$ Pa. Express the answer in terms of β and p_r, and also give its numerical value. Consider streambed sediments to be insulated by impermeable clays of low porosity. The depth of deposits H can be chosen arbitrarily.

As already mentioned, the condition is that reservoir pressure is caused by fluid compression relative to normal conditions. If a fluid is under normal atmospheric pressure, its fixed amount of substance occupies the largest possible volume, which means that the largest proportion of displaced fluid is achieved if in the final state it is at the lowest possible pressure. This can be realized if the depth of deposits H is directed to zero. We have:

$$\alpha = \frac{dV}{V} \approx \beta \Delta p.$$

Since the reservoir pressure is many times higher than atmospheric pressure, we find:

Answer:

$$\alpha_{max} \approx \beta p_{\mathrm{m.p.}} \approx 1.25\%.$$

A4 — 0.30 pts

Given the same data, estimate the maximum possible oil recovery factor α_{max} in the flowing mode, if there is water in the reservoir sediments below with a volume of kV_0 ($k = 9$) with initial oil reserves V_0. Consider the compressibility of water equal to the compressibility of oil. Express the answer in terms of β and p_r, and also give its numerical value. Consider that the liquid is taken from above, i.e. Only oil is taken. The depth of the deposits H can be chosen arbitrarily.

Similar to the previous point, the depth of deposits H should tend to zero. However, in this case, the oil recovery factor increases significantly, since not only the fluid, but also water is compressed, and since only oil is taken, its volume taken dV consists of the change in the volumes of oil and water::

$$\alpha = \frac{dV}{V_0} = \frac{dV_{\mathrm{o}} + dV_{\mathrm{w}}}{V_0} = (1+9)\beta\Delta p.$$

Thus:

Answer:

$$\alpha_{max} = 10\beta p_{\mathrm{m.p.}} \approx 12.5\%.$$

B1 — 1.00 pts

Let us consider the horizontal flow of fluid along the x axis between two parallel planes of height h. Distance between planes $w \ll h$. Determine the volumetric flow rate (hereinafter in all points of the problem - flow) of the liquid Q through the cross section wh. Express your answer in terms of η, w, h and the pressure gradient $dp(x)/dx$.

We obtain the distribution of fluid velocities along the coordinate z, directed perpendicular to the direction of flow from one plate to another. The beginning of the axis z is located in the middle between the plates. From the condition of conservation of momentum we have:

$$-hdzdp + \eta hdx\left(\frac{dv(z+dz)}{dz} - \frac{dv(z)}{dz}\right) = 0,$$

from which:

$$\frac{d^2v}{dz^2} = \frac{1}{\eta}\frac{dp}{dx}.$$

Integrate one times, find:

$$\frac{dv}{dz} = \frac{z}{\eta}\frac{dp}{dx} + A.$$

Since the velocity in the middle between the plates has an extremum $-$ $A = 0$. Integrating again, we find:

$$v(z) = C + \frac{z^2}{2\eta}\frac{dp}{dx}.$$

For $z = \pm w/2$ the velocity must vanish, from which find:

$$v(z) = -\frac{1}{2\eta}\frac{dp}{dx}\left(\frac{w^2}{4} - z^2\right).$$

For flux/flow Q have:

$$Q = \int\limits_{-w/2}^{w/2} v(z)hdz = -\frac{h}{2\eta}\frac{dp}{dx}\int\limits_{-w/2}^{w/2}\left(\frac{w^2}{4} - z^2\right)dz.$$

Integrating, we find:

Answer:

$$Q = -\frac{w^3 h}{12\eta}\frac{dp}{dx}.$$

B2 — 1.00 pts

An excess pressure Δp is created in the center of the gap. Find the dependence of the excess pressure p' in the slot on the coordinate x. Express your answer using Δp, Q, E, h, η and x.

Since the fluid flow spreads equally in two different directions, in each of them it is equal to $Q/2$. Let the x axis originate at the center of the slit and be directed along one of the flows. Let's use the result of point B1:

$$\frac{Q}{2} = -\frac{w^3 h}{12\eta}\frac{dp'}{dx}.$$

Using the empirical relation for the slit width, we obtain:

$$\frac{Q}{2} = -\frac{h^4 p'^3}{12\eta E^3}\frac{dp'}{dx}.$$

Integrate obtain expression:

$$\int\limits_{\Delta p}^{p'(x)} p'^3 dp' = \frac{p'^4(x) - \Delta p^4}{4} = -\frac{6Q\eta E^3 x}{h^4}.$$

Thus:

Answer:

$$p'(x) = \sqrt[4]{\Delta p^4 - \frac{24Q\eta E^3 x}{h^4}}.$$

B3 — 0.20 pts

The crack ends at a position corresponding to zero overpressure. Determine the crack length L. Express your answer in terms of Δp, E, h, η and Q.

At the maximum length of the crack, the pressure p' becomes zero, whence the half-length of the crack $L/2$ is:

$$\frac{L}{2} = \frac{\Delta p^4 h^4}{24Q\eta E^3}.$$

Thus:

Answer:

$$L = \frac{\Delta p^4 h^4}{12Q\eta E^3}.$$

B4 — 0.70 pts

Determine the crack volume V. Express your answer in terms of Δp, h, η, Q and E.

The volume of a crack is equal to twice the volume of its half:

$$V = 2\int\limits_0^L hw(x)dx = 2\int\limits_0^L \frac{h^2 p'}{E}dx.$$

Use expression for $p'(x)$:

$$V = \frac{2h^2\Delta p}{E}\int\limits_0^L \sqrt[4]{1 - \frac{24Q\eta E^3 x}{h^4 \Delta p^4}}dx = \frac{h^6 \Delta p^5}{12Q\eta E^4}\int\limits_0^1 \sqrt[4]{1-z}dz.$$

After elementary integration we find:

Answer:

$$V = \frac{h^6 \Delta p^5}{15Q\eta E^4}.$$

B5 — 0.30 pts

Calculate the maximum possible values of the crack length L_{max} and its volume V_{max}.

Answer:

$$L_{max} = \approx 83 \text{ m}.$$

C1 — 1.00 pts

Determine the speed v of motion of the fluid boundary when the front moves by an amount S. Express your answer in terms of p_1, p_2, L, η_1, η_2, k_1 and k_2.

Darcy's law within the framework of the problem can be rewritten in the following form:

$$v = -\frac{k}{\eta}\frac{\partial p}{\partial x}.$$

Velocity motion v identical for both liquid and constant by all length L. Then for the pressure gradient in the oil and in the water, taking into account the equality of their permeabilities $k_1 = k_2 = k$ obtain:

$$\frac{\partial p_{\text{o}}}{\partial x} = -\frac{\eta_1 v}{k_1} \qquad \frac{\partial p_{\text{in}}}{\partial x} = -\frac{\eta_2 v}{k_2}.$$

For total change pressure find:

$$p_2 - p_1 = \frac{\eta_2 S v}{k_2} + \frac{\eta_1 (L-S) v}{k_1}.$$

Thus:

Answer:

$$v = \frac{p_2 - p_1}{\dfrac{\eta_1 L}{k_1} + \left(\dfrac{\eta_2}{k_2} - \dfrac{\eta_1}{k_1}\right) S}.$$

C2 — 0.90 pts

Determine the dependence of the movement S of the front on time t. Express your answer using p_1, p_2, L, η_1, η_2, k and t

Since $v = dx/dt$ and $k_1 = k_2 = k$, we have:

$$dt = \frac{(\eta_1 L + (\eta_2 - \eta_1) S) dS}{k(p_2 - p_1)}.$$

Integrate, find:

$$t = \frac{1}{k(p_2 - p_1)} \int_0^S (\eta_1 L + (\eta_2 - \eta_1) x) dx = \frac{1}{k(p_2 - p_1)} \left(\eta_1 L S + \frac{(\eta_2 - \eta_1) S^2}{2}\right).$$

Bringing the quadratic equation to classical form:

$$S^2 - \frac{2\eta_1 L S}{\eta_1 - \eta_2} + \frac{2k(p_2 - p_1)t}{\eta_1 - \eta_2} = 0.$$

Solve, obtain:

$$S(t) = \frac{\eta_1 L}{\eta_1 - \eta_2} \pm \sqrt{\left(\frac{\eta_1 L}{\eta_1 - \eta_2}\right)^2 - \frac{2k(p_2 - p_1)t}{\eta_1 - \eta_2}}.$$

Since $x(0) = 0$ – let's choose the root with a minus sign. Finally:

Answer:

$$S(t) = \frac{\eta_1 L}{\eta_1 - \eta_2} - \sqrt{\left(\frac{\eta_1 L}{\eta_1 - \eta_2}\right)^2 - \frac{2k(p_2 - p_1)t}{\eta_1 - \eta_2}}.$$

C3 — 0.50 pts

Determine the total time τ for oil displacement from the field. Express the answer in terms of p_1, p_2, L, η_1, η_2 and k and calculate it.

The total time τ of oil displacement is $t(L)$. Thus:

Answer:

$$\tau = \frac{(\eta_1+\eta_2)L^2}{2k(p_2-p_1)} \approx 26 \text{ year}.$$

C4 — 0.80 pts

Under what condition on the parameters of the system will the motion of the boundary be stable, that is, if the shape of the boundary deviates small from flat, this deviation will not increase? Write the stability condition in terms of η_1, η_2, k_1 and k_2. Is the fluid flow considered in paragraphs C2 *and* C3 *stable?*

The deviation will not increase if $v(x+dx)<v(x)$. Note that the numerator of the expression for speed is constant, and the denominator is a linear function of x with slope $a = \eta_2/k_2 - \eta_1/k_1$. The magnitude of the speed decreases with increasing x, if the value a is positive, i.e., provided:

Answer:

$$\frac{\eta_2}{k_2} > \frac{\eta_1}{k_1}.$$

D1 — 0.80 pts

Find the dependence of the fluid flow velocity in such a pipe on the distance to the pipe axis $v(r)$, the maximum value of the velocity v_{max} and the total fluid flow Q through the cross section of the cylinder. Express your answers using Δp, η, L, R and r.

Let us select a cylinder with radius r and length L. Let us write down the condition for the constancy of its momentum:

$$\pi r^2 \Delta p + 2\pi r L \eta \frac{\partial v}{\partial r} = 0 \Rightarrow \frac{\partial v}{\partial r} = -\frac{r}{2}\frac{\Delta p}{\eta L}.$$

Integrate, obtain:

$$v = C - \frac{\Delta p r^2}{4\eta L}.$$

Since $v(R) = 0$, we obtain:

For flow Q we have:

$$Q = \int_0^R v(r)\cdot 2\pi r dr = \frac{\pi \Delta p}{2\eta L}\int_0^R (R^2 - r^2) r dr.$$

Integrating, we find:

Answer:

$$v(r) = \frac{\Delta p (R^2 - r^2)}{4\eta L}.$$

D2 — 0.20 pts

Express the distribution of fluid velocity $v(r)$ in terms of the total flow Q, R and r.

We express the combination $\Delta p/(4\eta L)$ in terms of the full flow Q:

$$\frac{\Delta p}{4\eta L} = \frac{2Q}{\pi R^4}.$$

Substitute in expression for $v(r)$, we find:

Answer:

$$v(r) = \frac{2Q}{\pi R^2}\left(1 - \frac{r^2}{R^2}\right).$$

D3 — 0.20 pts

Find the flow Q in the face section at a distance h from its lower edge and the corresponding expression for the vertical velocity $v(r,h)$ depending on the distance to the axis r and the height h. Express your answers using Q_0, H, R, r and h.

Since the liquid enters the cylinder uniformly along its side surface, we have:

$$\frac{dQ}{dh} = \frac{Q_0}{H}.$$

Since $Q(0) = 0$, we find:

Let's use the expression obtained in step D2:

$$v(r,h) = \frac{2Q(h)}{\pi R^2}\left(1 - \frac{r^2}{R^2}\right).$$

Substitute $Q(h)$, we get:

Answer:

$$Q(h) = \frac{Q_0 h}{H}.$$

D4 — 0.30 pts

Consider a ring of height dh with inner and outer radii r and $r + dr$ respectively. Using the fact that the fluid is incompressible, show that from the condition that the volume of the fluid inside the selected ring is constant, the following relation follows:

$$\frac{\partial v}{\partial h} = -\frac{1}{r}\frac{\partial(u_r r)}{\partial r}.$$

You can use this relation even if you could not prove it.

Let us write the expression for the total fluid flow q into the ring:

$$q = 2\pi r dr v(r,h+dh) - 2\pi r dr v(r,h) + 2\pi dh(r + dr)u_r(r + dr,h) - 2\pi dh r u_r(r,h) = 0.$$

Thus:

$$2\pi r dr dv + 2\pi dr d(r u_r) = 0,$$

or same:

$$\frac{\partial v}{\partial h} + \frac{1}{r}\frac{\partial(r u_r)}{\partial r} = 0.$$

Which is what we needed to prove.

D5 — 0.50 pts

Find the radial velocity of the fluid flow $u_r(r,h)$ depending on the distance to the axis r and the height h, as well as the maximum value of its modulus $u_{r(max)}$. Express your answers using Q_0, R, H, h and r.

Consistently use the results of points D4 and D3:

$$\frac{\partial(r u_r)}{\partial r} = -r\frac{\partial v}{\partial h} = -\frac{2Q_0 r}{\pi R^2 H}\left(1 - \frac{r^2}{R^2}\right).$$

Note, that for $r = 0$ quantity $r u_r$ also equal zero. Temporarily relabel r as z in the integrand function and obtain:

$$ru_r(r,h) = -\frac{2Q_0}{\pi R^2 H}\int\limits_0^r \left(1 - \frac{z^2}{R^2}\right) zdz = -\frac{Q_0}{\pi R^2 H}\left(r^2 - \frac{r^4}{2R^2}\right).$$

Thus:

The maximum value of u_r is achieved when its derivative with respect to r is equal to zero:

$$\frac{du_r}{dr} = 0 \Rightarrow 1 - \frac{3r^2}{2R^2} = 0 \Rightarrow r_{max} = \sqrt{\frac{2}{3}}R.$$

Then for the maximum value of the radial velocity component u_{max} we find:

Answer:

$$u_r(r,h) = -\frac{Q_0}{\pi R^2 H}\left(r - \frac{r^3}{2R^2}\right).$$

D6

0.10 pts

What is the ratio $u_{r(max)}/v_{max}$? Express your answer in terms of R and H.

Answer:

$$\frac{u_{r(max)}}{v_{max}} = \frac{\sqrt{2}R}{3\sqrt{3}H}$$